

ST JAMES, MN, USA

WMO: 726554

Lat: 43.986N

Lon: 94.558W

Elev: 325

StdP: 97.48

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 54906

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.0	-21.1	-27.4	0.3	-23.0	-25.0	0.4	-20.8	14.6	-7.1	13.1	-8.9	3.8	300	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	32.4	23.5	31.0	22.8	29.0	21.9	26.2	29.3	24.9	28.8	23.7	27.4	5.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	21.2	28.3	23.7	19.3	27.3	22.5	17.9	26.2	83.8	28.8	77.6	28.9	72.8	27.8	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.5	11.1	9.8	DB	-27.0	35.1	2.9	1.6	-29.1	36.2	-30.8	37.2	-32.4	38.1	-34.5	39.2	(4)
(5)			WB	-27.3	27.9	2.8	1.6	-29.3	29.0	-31.0	29.9	-32.6	30.9	-34.6	32.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	-8.3	-6.1	0.4	8.2	15.2	20.5	22.4	20.9	17.3	9.8	2.0	-5.8	(6)	
	DBStd	12.19	7.10	6.95	6.86	5.69	5.00	3.75	3.04	3.08	4.68	5.32	6.15	6.71	(7)	
	HDD10.0	2239	566	452	304	97	11	0	0	0	4	69	247	489	(8)	
	HDD18.3	4108	824	685	554	306	123	21	4	11	74	268	489	748	(9)	
	CDD10.0	1556	0	0	8	44	173	314	384	339	222	64	8	0	(10)	
	CDD18.3	382	0	0	0	2	26	86	131	91	43	4	0	0	(11)	
	CDH23.3	3563	0	0	2	38	314	779	1218	729	417	63	2	0	(12)	
	CDH26.7	1114	0	0	0	9	117	263	392	192	126	16	0	0	(13)	
(14) Wind	WSAvg	4.1	4.8	4.8	4.6	4.8	4.8	3.8	2.8	2.5	3.3	4.2	4.6	4.7	(14)	
(15) Precipitation	PrecAvg	769	18	18	42	82	105	121	100	94	75	56	34	23	(15)	
	PrecMax	1151	37	45	79	165	173	260	195	250	222	166	103	61	(16)	
	PrecMin	516	2	1	2	25	33	59	17	18	9	8	2	3	(17)	
	PrecStd	155	10	11	22	36	41	48	45	52	58	40	26	14	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	12.6	22.2	27.7	33.0	33.9	33.1	32.4	32.4	28.8	21.3	12.3	(19)	
		MCWB	4.6	9.2	13.9	16.2	19.6	22.3	25.7	25.7	21.9	17.3	12.9	9.6	(20)	
	2%	DB	4.1	8.0	17.3	23.0	29.9	32.0	32.3	30.9	29.9	24.8	17.5	7.7	(21)	
		MCWB	2.0	5.9	12.2	13.2	18.0	21.8	25.4	25.0	21.3	15.6	11.6	4.8	(22)	
	5%	DB	2.5	4.9	13.0	21.0	27.2	29.2	30.3	28.7	27.8	21.9	13.9	4.2	(23)	
		MCWB	1.2	3.0	9.7	12.5	16.9	20.6	24.3	23.5	20.0	14.4	9.1	2.4	(24)	
	10%	DB	1.2	2.6	10.0	17.6	23.8	27.7	28.8	27.4	26.0	18.1	11.1	2.6	(25)	
		MCWB	0.3	1.4	7.6	11.3	15.5	20.0	23.4	22.3	18.8	12.9	7.6	1.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.2	9.9	16.2	18.2	23.2	26.1	28.7	28.0	24.6	21.6	15.0	9.7	(27)	
		MCDB	8.6	12.2	18.7	23.7	28.5	29.4	32.2	30.1	29.2	25.0	19.2	11.8	(28)	
	2%	WB	2.6	5.9	12.8	15.7	20.1	24.2	26.6	26.1	23.0	17.5	12.2	5.1	(29)	
		MCDB	4.1	8.1	16.1	21.1	26.4	29.1	29.8	28.7	27.4	20.6	16.6	7.6	(30)	
	5%	WB	1.4	3.2	10.2	13.7	18.6	22.7	25.2	24.4	21.6	15.6	9.9	3.0	(31)	
		MCDB	2.5	4.6	13.1	18.8	24.2	26.9	29.3	27.5	26.3	20.1	13.7	4.4	(32)	
	10%	WB	0.2	1.6	7.4	11.9	17.1	21.4	24.1	23.0	20.1	13.7	7.5	1.5	(33)	
		MCDB	1.1	2.6	9.4	16.7	22.1	25.9	27.8	26.1	24.2	18.2	10.0	2.6	(34)	
(35) Mean Daily Temperature Range	MDBR		8.7	8.9	8.9	11.1	11.4	10.5	10.8	10.9	12.6	11.6	9.9	8.3	(35)	
	5% DB	MCDBR	9.7	10.7	14.0	16.7	15.5	12.9	11.6	11.9	14.5	16.0	14.8	10.9	(36)	
		MCWBR	8.1	8.5	9.6	9.0	6.9	5.9	6.7	7.0	7.0	8.2	9.1	8.7	(37)	
	5% WB	MCDBR	9.0	9.9	12.9	13.8	12.5	10.8	10.6	10.4	12.4	12.4	13.6	10.2	(38)	
		MCWBR	7.6	8.0	9.4	8.6	6.7	6.4	6.9	7.1	7.0	7.9	9.2	8.3	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.266	0.284	0.310	0.363	0.383	0.403	0.415	0.411	0.369	0.325	0.285	0.264	(40)	
	taud		2.453	2.406	2.451	2.337	2.347	2.341	2.293	2.286	2.376	2.496	2.528	2.474	(41)	
	Ebn at Noon		878	916	933	900	888	868	852	842	854	857	848	844	(42)	
	Edh at Noon		71	89	98	119	122	123	128	124	105	81	66	63	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.83	2.87	3.78	4.65	5.44	6.04	6.49	5.37	4.28	2.82	1.86	1.38	(44)	
	RadStd		0.20	0.33	0.38	0.45	0.40	0.47	0.26	0.40	0.29	0.37	0.16	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (65 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page